MACA: A Relative Radiometric Correction Method for Multiflight Unmanned Aerial Vehicle Images Based on Concurrent Satellite Imagery

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***Abstract*— Unmanned aerial vehicle (UAV) offers an unprecedented observing potential with ultrahigh spatial and temporal resolutions and high flexibility. However, it remains difficult to solve the radiometric inconsistency of multiflight UAV imagery. This study proposed a relative radiometric correction method for multiflight UAV images based on concurrent satellite imagery (MACA) consisting of two steps, i.e., cross-sensor spectral fitting (CSF) and fine-resolution spectral calibration (FSC). In CSF, relationships of multiflight UAV reflectance and the concurrent satellite reflectance were established for generating the fine-resolution reference imagery. Subsequently, a relative radiometric correction model was constructed in the FSC step to correct multiflight UAV imagery. The performance of MACA was evaluated using multiflight UAV datasets acquired on six cropland sites and a concurrent Sentinel-2 image. Compared with four typical or state-of-the-art relative correction methods, the correction using MACA yielded better consistency between UAV and Sentinel-2 data, regardless of individual spectral bands (R^2 = 0.79–0.86, root mean square error (RMSE) = 0.004–0.019) or vegetation indices (VIs) (R^2 = 0.80–0.86, RMSE = 0.024–0.054). Moreover, the prediction of plant nitrogen accumulation (PNA) based on the MACA-corrected UAV data had the highest accuracy and showed the spatial variation most significantly within and between fields for all sites. The results demonstrated that MACA was more robust in reducing spectral mismatch across sensors and eliminating the subjective error of pseudo-invariant features (PIFs) selection. MACA has the potential to be used to cross-calibrate multisensor data into a consistent standard, which will benefit multisensor synergies.**

***Index Terms*— Cubist, machine learning, nitrogen, radiometric correction, Sentinel-2, unmanned aerial vehicles (UAVs).**

I. Introduction

In THE past few decades, unmanned aerial vehicle (UAV) has become a game changer in remote sensing, providing data of unprecedented spatial and temporal resolutions and with high flexibility, high efficiency, and low cost [1]. UAV-based remote sensing (UAV-RS) has widely been used in civilian fields, such as precision agriculture, disaster management, geographic mapping, and atmospheric monitoring [2], [3]. The applications of UAV-RS seem to be unlimited and booming. However, due to low flight durations and limited flight altitudes, the coverage of a UAV flight is generally small (a few km^2), which hinders its potential development. To map a large area at a high spatial resolution, multiflight UAV images need to be acquired [4], [5]. Unfortunately, similar ground objects between images acquired in different flights may express different spectral characteristics due to variation in acquisition conditions (e.g., status or posture of a sensor, solar illuminance, and weather) [6]. This problem cannot be ignored, because the comparative analysis or comprehensive application of inconsistent spectral information from multiflight UAV imagery can lead to misleading findings with large uncertainties [2]. Therefore, a robust radiometric correction method is required to ensure high-quality and reliable UAV data.

In most cases, radiometric correction of UAV imagery uses the ground targets with known reflectance (e.g., calibration panels) as references in the field [7], [8]. The relationship between the reflectance of ground targets and the corresponding digital numbers (DNs) from a UAV-mounted sensor is built to calibrate all imagery DNs to surface reflectance [9]. However, the relationship has been empirically assumed to be linear [10], [11], exponential [12], logarithmic [13], or second-degree polynomial [14], varying with sensor type, land cover, and acquisition conditions. This type of radiometric calibration can only be used to adjust the scale of measured values by relating radiance values to the solar irradiation at the moment when the image of the calibrated reflectance panel was collected [10]. Moreover, the selection of ground targets tends to be subjective, which would introduce uncertainty into the empirical relationships. Hence, these approaches using ground targets are not robust. These potential limitations pose a serious threat to the applications of UAV-RS, especially multiflight UAV imagery. Therefore, further radiometric correction (or radiometric normalization)

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is needed to eliminate the radiometric discrepancy between multiflight UAV images caused by acquisition conditions and to reveal actual spectral response of ground objects.

Radiometric correction can be categorized into two types: absolute radiometric correction and relative radiometric normalization (RRN) [15]. Absolute radiometric correction, also referred to as absolute atmospheric correction, aims to completely remove atmospheric effects based on a single image [16], [17]. However, it requires a large amount of ancillary data (air temperature, atmospheric pressure, relative humidity, etc.) at the time of remote sensing imagery collection to accurately estimate atmospheric effects, which is difficult even through complex computations [18]–[20]. In contrast, RRN uses a reference image to normalize images without other ancillary data, and it is computationally simple [21], [22]. Thus, RRN has widely been used to increase the radiometric agreement between remote sensing images acquired at different times [23], [24].

RRN has already applied to satellite and airborne data [25]–[27], but RRN for UAV data is more specific and barely reported. According to the reference data source, RRN can be further divided into two categories: interior and exterior reference methods. Interior reference methods select one image as a reference and normalize other images to the same radiometric scale as the reference image [28]–[30]. This is based on the assumptions that adequate pseudo-invariant features (PIFs) can be detected in both reference and other images, and that the radiometric differences between images can be expressed by a linear or nonlinear relationship [21]. Generally, interior reference methods are used for radiometric normalization between multitemporal satellite images acquired from the same sensor and the same area [28], [31]. Particularly, Rahman *et al.* [32] applied interior reference methods to multiflight airborne imagery with the overlapping area of ~9.45 km^2. However, the UAV images acquired in different flight missions cannot rival the overlapping extent of satellite and airborne imagery. Hence, interior reference methods might not be appropriate for the radiometric correction of UAV images with insufficient PIFs (e.g., multiflight UAV images with partial overlapping areas or without overlapping areas). The other category of RRN is exterior reference methods, which radiometrically correct images from one data source by employing concurrent images from another data source (usually high-quality and lower resolution data) as [33], [34]. Exterior reference methods are mainly used to cross-calibrate satellite data from different instruments. For example, Houborg and McCabe [35] used Landsat and MODIS to enhance radiometric consistency of CubeSat data. Since satellite-based systems have been rigorously calibrated and validated (e.g., MODIS, Landsat, and Sentinel-2), they are potential exterior references for the radiometric correction of UAV imagery. While satellite data have been used to calibrate UAV-based retrieval models [2], [3], no study has investigated the radiometric correction of multiflight UAV images based on satellite data.

In this study, the RRN concept was adopted to enhance the radiometric consistency of multiflight UAV imagery. We proposed a relative radiometric correction method for multiflight UAV images based on concurrent satellite imagery

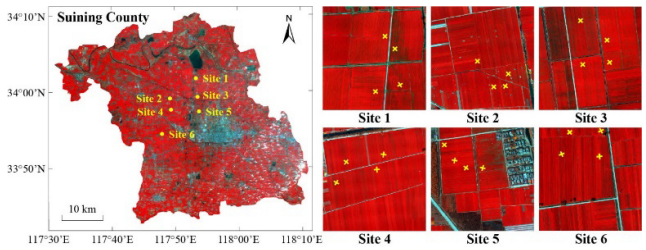


Fig. 1. Study sites in Suining, Jiangsu, China. The county is covered by a Sentinel-2 image at 10 m. The coverage for each site presents the UAV image with the spatial resolution of 0.1 m. All images are displayed as false color composites using the near infrared (NIR), red, and green bands. The crosses display the distribution of ground sampling areas at each site.

(MACA). Its performance on the radiometric correction of multiflight UAV images was evaluated and compared with four traditional or state-of-the-art methods, i.e., the mean of ratio (MOR), the PIF, multivariate alteration detection-based RRN (MAD-RRN), and support vector machine regression-based RRN (SVR-RRN). To do this, we collected multiflight UAV images at six sites and acquired the concurrent Sentinel-2 data as the exterior reference. After the radiometric correction of multiflight UAV imagery based on Sentinel-2 data, we evaluated the performance of the three radiometric correction methods in individual spectral bands and vegetation indices (VIs). Moreover, we analyzed the contributions of the radiometric correction methods to the predictions of winter wheat nitrogen accumulation over the six sites. Finally, we discussed the potential of the MACA method for enhancing multisensor synergies.

II. Materials and Methods

The analysis presented herein relied on the integration of remotely sensed imagery from UAV-based multispectral camera and Sentinel-2 acquired near coincidently in April 2019 with a temporal gap of 0–1 day (see Table I). During the data acquisition period, the sky conditions were clear and the land cover remained unchanged. Therefore, the Sentinel-2 and UAV imagery were considered as concurrent image pairs.

A. Study Area

The study area is located in the Yangtze-Huai plain, one of the major crop production regions in China (see Fig. 1). The major crops in this region are winter wheat and rice, which were planted in a rotation cropping system. Winter wheat was the main crop at these sites during the data acquisition time (i.e., April). The selection of study sites was based on the following criterion: 1) the sites were mainly covered by winter wheat fields with variation in growth status and spatial heterogeneity; 2) the sites had adequate PIFs detectable in both UAV and satellite images for radiometric correction analysis; and 3) the sites were discretely distributed across a satellite image. The area of 400 m × 400 m for each site contains 16 million UAV pixels at 0.1 m or 1600 Sentinel-2 pixels at 10 m, which is sufficient for model training and validation given the available resources for field campaigns.

B. Data Acquisition

1) UAV Data: The UAV platform was a DJI Matrice 600 Pro hexarotor (DJI Innovations, Shenzhen, China), includ-

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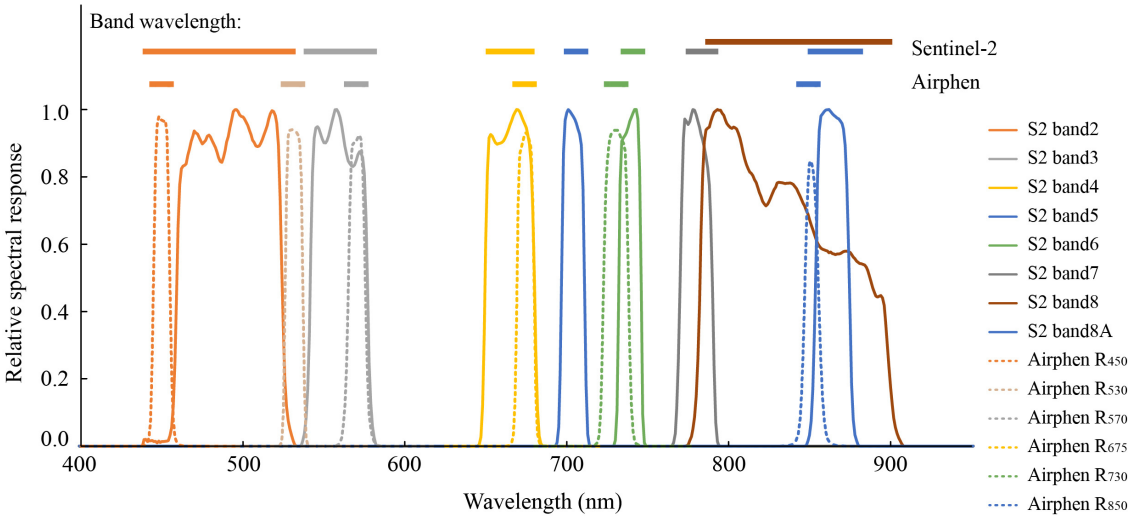


Fig. 2. Relative spectral responses of the Sentinel-2 MSI sensor and the UAV-mounted Airphen camera.

ing global positioning system (GPS), flight controller, DJI lightbridge, and expansion bay. Its maximum payload capacity was 6 kg. The maximum hovering time varied from 16 to 38 min depending on the payload. An Airphen multispectral camera (Hiphen, Avignon, France) was mounted on the UAV platform. Airphen had six bands with a spectral resolution of 10 nm (see Table I). The UAV was flown at 200-m altitude with the Airphen camera at a speed of 10 m/s. The multispectral images were collected once every second with 85% forward and side overlaps (1 photograph/s). The six campaigns of UAV data collection were conducted between 10:00 and 14:00 under cloud-free conditions. For the UAV campaigns, six to nine ground control points (GCPs) were placed within each site for geometric correction and their coordinates were measured using Trimble GeoXH6000 (Trimble, CA, USA). Before each flight, a standard panel (with different reflectances across bands) from Hiphen was placed within the UAV flight overpass for reflectance calibration. All UAV images were processed in Agisoft Photoscan Professional 1.4.5 (Agisoft LLC, St. Petersburg, Russia) with the Airphen camera plug-in, including reflectance calibration, image alignment and mosaicking, dense point clouds and mesh generation, and orthomosaic production [36], [37]. The pixel size of the orthomosaics was 0.1 m.

2) Sentinel-2 Data: The Sentinel-2 multispectral instrument level 2A (L2A) images were acquired from the Copernicus Open Access Hub (https://scihub.copernicus.eu/). The L2A product is bottom-of-atmosphere reflectance in cartographic geometry, which is atmospherically corrected using the atmospheric correction (S2AC) processor [38] based on the library for radiative transfer (LIBRADTRAN) radiative transfer model [39] and ready to use [40]. The principles and details of Sentinel-2 radiometric calibration and products validation are fully described in [41], reporting less than 3% uncertainty in the reflective wavelength calibration. Due to the rigorously radiometric calibration and validation, the Sentinel-2 L2A product is used as the reference data in MACA.

All L2A bands were resampled to 10 m using the super resolution function in the Sentinel Application Platform (SNAP) provided by European Space Agency (ESA). Only the five bands of Sentinel-2 with similar wavelength configurations to the Airphen bands were considered to reduce the uncertainty caused by the spectral mismatch between Sentinel-2 and UAV data (see Fig. 2 and Table I). The reflectance images from Sentinel-2 and UAV were co-registered based on manually selected GCPs and cropped to the same area in ENVI 5.1 (Exelis, Boulder, CO, USA).

TABLE I
BASIC INFORMATION OF THE UAV AND SENTINEL-2 DATA USED IN THIS STUDY

	UAV (Airphen)	Sentinel-2
Spectral bands (band number: central wavelength)		
Blue	1: 450 nm	2: 490 nm
Green	2: 530 nm	
	3: 570 nm	3: 560 nm
Red	4: 675 nm	4: 665 nm
Red edge (RE)	5: 730 nm	6: 740 nm
Near Infrared (NIR)	6: 850 nm	8A: 865 nm
Pixel size	0.1 m	10 m
Acquisition date	April 6 - 7, 2019	April 7, 2019

3) Agronomic Measurements: Nitrogen (N) is a component of many important compounds in plants and plays a key role in plant growth [42], [43]. The estimation of plant nitrogen accumulation (PNA) is useful for the diagnosis of N status in crop production and grain quality. In agricultural demonstration bases, multiflight UAV images are often required for monitoring the PNA and further making N fertilization recommendations. As such, we used PNA as a typical application scenario of MACA in this study.

Ground destructive samplings for the measurements of PNA in winter wheat were taken on the same dates as the UAV campaigns (see Table I). Each site encompassed four randomly selected homogeneous areas of 10 m × 10 m (see Fig. 1), corresponding to the pixel size of the Sentinel-2 imagery. Three subareas of 0.5 m × 0.5 m diagonally distributed

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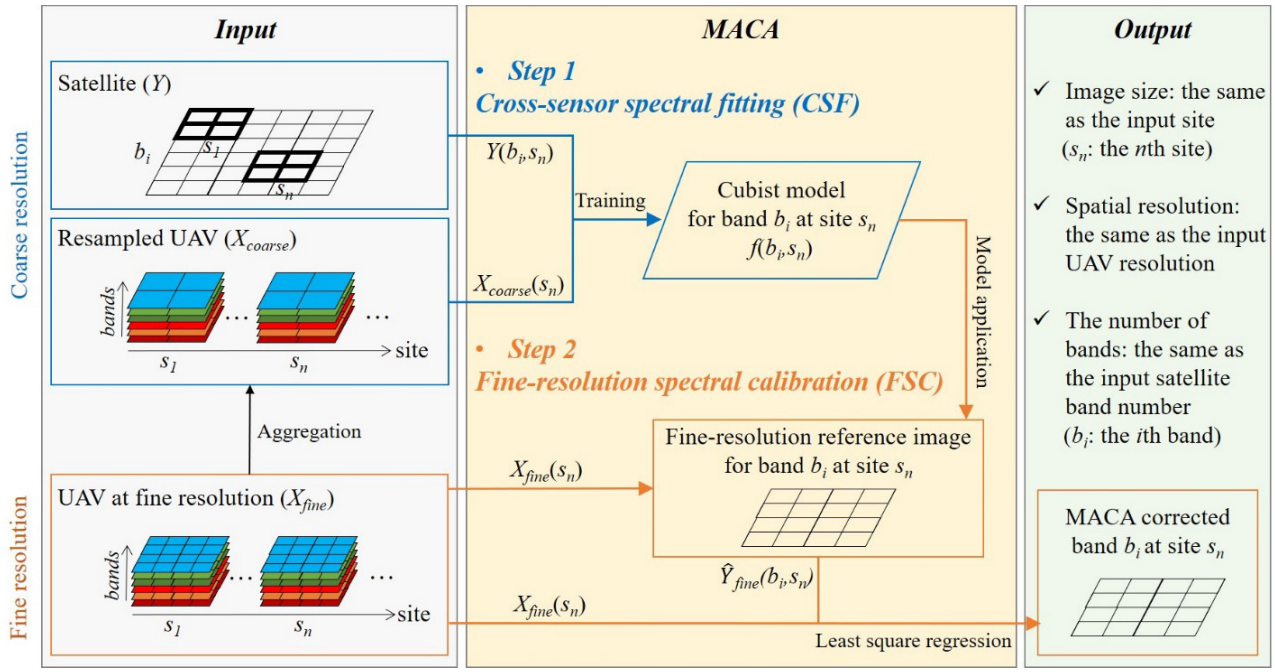


Fig. 3. Technical flowchart of the MACA method.

within each homogeneous area were selected as the sampling plots. The central coordinate of each plot was collected using Trimble GeoXH6000 at subcentimeter positional accuracy. For each plot, the number of tillers was recorded and 20 hills of wheat plants were randomly cut above the ground surface and separated into leaves, stems, and ears. All samples were oven-dried at 105 °C for 30 min and later at 80 °C for 48 h to a constant weight. The dried samples were weighed to measure the above ground biomass (AGB), which is the sum of dry weights of aboveground leaves, stems, and ears [44]. According to the number of tillers, the AGB per plot was calculated. To measure the N concentration in wheat organs (i.e., leaves, stems, and ears), the samples were grounded and stored in labeled bags for chemical analysis. The N concentration content by organ was determined using the micro-Keldjahl method [45] with SEAL AutoAnalyzer 3 HR (SEAL Analytical, Ltd., Germany). Finally, the PNA (kg/ha) was calculated as the product of N concentration (%) and dry biomass (kg/ha) of whole aboveground plants

$$\mathrm{PNA} = \sum_{i=1}^{3} (N_i \times \mathrm{AGB}_i) \tag{1}$$

where i is the wheat organ, including leaves, stems, and ears.

C. MACA

The MACA was proposed as a relative radiometric correction method to radiometrically normalize multiflight UAV data into Sentinel-2 consistent surface reflectance. MACA consists of two main steps: 1) cross-sensor spectral fitting (CSF) and 2) fine-resolution spectral calibration (FSC) (see Fig. 3). Before the implementation of MACA, all input images should be reprojected to the same coordinate system if they are not. To preserve the spectral integrity of the original UAV images, pixel aggregation was used to resample UAV orthomosaics from 0.1 m to 10 m. A regression model was established to relate the resampled UAV data to the target Sentinel-2 surface reflectance data and then applied to the original UAV imagery at 0.1 m to create the reference image at the fine resolution. Based on the reference image, a relative radiometric correction model was constructed to generate the radiometrically corrected UAV image at fine resolution.

1) Cross-Sensor Spectral Fitting: Cubist was used to establish the relationship between the resampled UAV data and Sentinel-2 data. It (RuleQuest; www.rulequest.com) is a rule-based regression model and represents a nonparametric machine-learning technique. Cubist creates a series of "if-after-after" rules and adopts Quinlan's M5 model tree [46]–[48] to learn the relationship between a training dataset of input covariates and the target variable. A tree grows and each terminal branch on the tree is represented by a multivariate linear regression model associated with a specific rule. As long as the subset of covariates satisfies the rules, the corresponding model will be used for prediction. The complexity of the tree is determined by the characteristics of the training dataset and the number of allowed rules (*nrul*). For the details of the cubist regression algorithm, readers are referred to relevant references [49], [50].

Training models were built for each site using the target (Sentinel-2) and input covariates (the resampled UAV) data of the full scene domain (see Fig. 3). A total of 30 models (5 band-specific models × 6 sites) were built. The five band-specific prediction models correspond to Sentinel-2 band 2 (blue), 3 (green), 4 (red), 6 (RE), and 8A (NIR), respectively. For each model, all bands of the resampled UAV data were used as input covariates. The number of rules (*nrul*) is required to be optimized for cubist models. According to previous studies [35], [51], [52] and our trail-and-error experiments,

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the *nrul* was initially set from 1 to 20 with the step of 1 and automatically optimized by a cost function defined as the root mean square error (RMSE) of a training subset. The optimal number of maximum rules was then used to drive the regression model. Consequently, a total of 30 prediction models (i.e., five band-specific prediction models for each of six study sites) were constructed by cubist.

2) Fine-Resolution Spectral Calibration: Those cubist models were applied to the corresponding UAV data at 0.1 m to generate fine-resolution reference images. Based on the reference images, the concept of RRN was adopted to enhance the radiometric consistency of multiflight UAV imagery by calibrating spectral information at fine resolution. Least squares regression was used to construct the relative radiometric correction models for each band by minimizing the sum of the squares of the residuals

$$\min = \sum_{p=1}^{n} (y_i - a_i x_i - b_i)^2 \tag{2}$$

where x_i and y_i represent the ith band of the original UAV orthomosaic and the reference image, respectively. p and n are the pth pixel and the number of input pixels, respectively. To solve (2), the normalized coefficients (a_i and b_i) can be obtained as follows:

$$a_i = \frac{S_{x_i y_i}}{S_{x_i x_i}}, \quad b_i = \bar{y}_i - a_i \bar{x}_i \tag{3}$$

with

$$S_{x_i x_i} = \frac{1}{n} \sum_{p=1}^{n} (x_i - \bar{x}_i)^2 \tag{4}$$

$$S_{x_i y_i} = \frac{1}{n} (x_i - \bar{x}_i)(y_i - \bar{y}_i) \tag{5}$$

where $S_{x_i x_i}$ and $S_{x_i y_i}$ represent the variance of the original UAV data and the covariance between the original UAV and the reference data, respectively. Based on the corresponding radiometric correction models, the original UAV orthomosaics were translated into Sentinel-2 consistent estimates at 0.1-m resolution.

D. Comparison and Evaluation

Four typical or state-of-the-art RRN approaches, i.e., MOR, PIF, MAD-RRN, and SVR-RRN, were considered as the benchmark and compared with MACA. All methods used the Sentinel-2 surface reflectance data as the reference as MACA did.

MOR [6] and PIF [28] are two widely used methods for the relative radiometric correction of multisource satellite data and can be implemented for UAV and satellite image pairs [21], [53]–[55]. MOR is based on the assumption that the cross-sensor relationship is a simple ratio. Note that only the Airphen bands with similar central wavelengths to Sentinel-2 were used in MOR and PIF, namely, excluding band 2 (see Table I). The ratios of reflectance for each pair of Sentinel-2 and the resampled 10-m UAV bands were calculated. Then, the reflectance of the corresponding band of 0.1-m UAV imagery was corrected by multiplying the mean of the ratios. While the MOR method uses all pixels within the image scene and assumes a cross-sensor ratio relationship, the PIF method usually conducts a linear relationship of cross-sensor data based on selected PIFs [28]. We selected more than 45 homogenous regions of interest (ROIs), including winter wheat, roads, and other land cover types for each site (15–20 ROIs for each class). The values of reflectance in each ROI were averaged to 10 m for corresponding to the Sentinel-2 pixel size. For each band, a linear model between the reflectance of features from UAV imagery and the corresponding reflectance of Sentinel-2 features was established and applied to the whole UAV imagery of each site. Based on the PIF theory, MAD-RRN [56], [57] and SVR-RRN [58] are the modified PIF-based RRN methods with automatically selecting invariant pixels. While MAD-RRN uses the linear regression as the PIF method, SVR-RRN employs SVR for fitting the spectral relationship of the selected invariant pixels between reference and target imagery.

To evaluate the performance of different radiometric correction methods, the UAV images radiometrically corrected by different methods were resampled to 10 m and compared with Sentinel-2 imagery, including the corresponding spectral bands and six commonly used VIs (see Table II), using RMSE and the coefficient of determinations (R^2).

To evaluate the radiometric correction methods in quantitative application, the PNA of winter wheat was predicted based on multiflight UAV data before and after radiometric corrections. The VIs identified in Table II were extracted from UAV imagery for each plot and used to build PNA prediction models. The correlations between the measured PNA and the VIs from different UAV imagery (before and after correction by different correction methods) were analyzed using the Pearson correlation coefficient. The random forest approach was used to predict PNA, since it has been shown to provide a fast and accurate estimation of crop traits [65], [66]. Six random forest models were developed using a training dataset based on: 1) uncorrected UAV imagery; 2) MOR-corrected UAV imagery; 3) PIF-corrected UAV imagery; 4) MACA-corrected UAV imagery; 5) MAD-RRN-corrected UAV imagery; and 6) SVR-RRN-corrected UAV imagery. For all random forest models, the number of variables available for splitting at each tree node (*mtry*) was set as default. A total of 1000 decision trees (*ntree*) were used for all models to ensure stable predictions and computational efficiency. Training sets were compiled by randomly sampling 2 out of 3 plots for each homogenous area (i.e., approximately 2/3 of the entire dataset), with the remainder used as the test set. The accuracies of the random forest models were evaluated using R^2 and RMSE. The random forest models in all cases were implemented in Python.

III. Results

A. Evaluation of Correction for Each Band

Fig. 4 shows the correlations of Sentinel-2 surface reflectance and the corresponding reflectance of UAV imagery for the six study sites before and after radiometric correction. Clearly, the proposed MACA achieved the best accuracy. The correction using MACA yielded an increase in R^2

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TABLE II
VIs USED IN THIS STUDY

Vegetation index		Formulation [a]	Reference
Abbreviation	Full name		
NDVI	Normalized difference vegetation index	(NIR-R)/(NIR+R)	[59]
CIgreen	Green chlorophyll index	(NIR/G)-1	[60]
CIre	Chlorophyll index red-edge	(NIR/RE)-1	[61]
MTCI	MERIS terrestrial chlorophyll index	(NIR-RE)/(RE-R)	[62]
ENDVI	Enhanced NDVI	(NIR+G-2×B)/(NIR+G+2×B)	[63]
EVI	Enhanced vegetation index	2.5×(NIR-R)/(NIR+6.0×R-7.5×B+1)	[64]

[a] B, G, R, RE, and NIR represent the blue, green, red, red-edge, and near infrared bands, respectively.

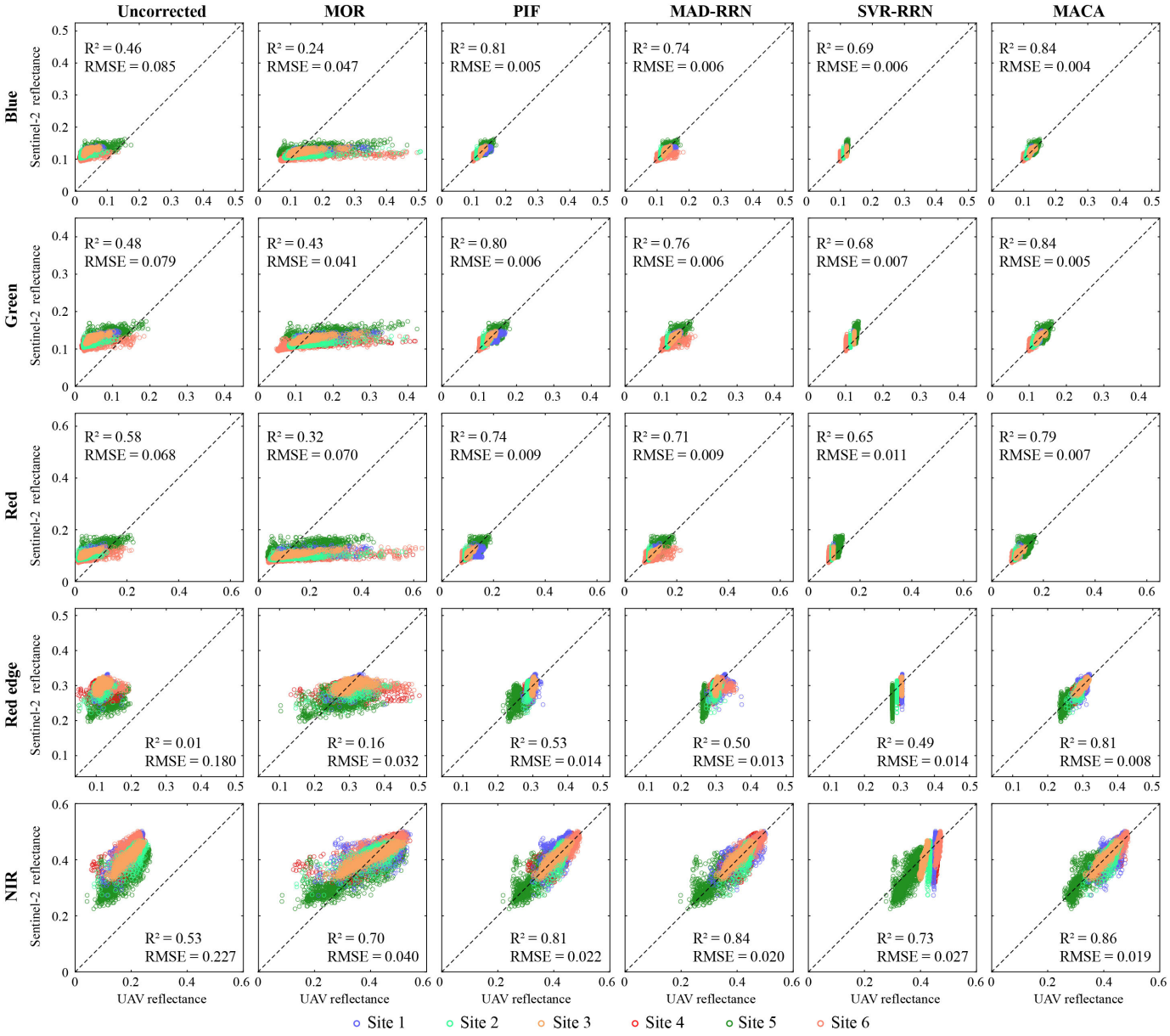


Fig. 4. Scatterplots of the Sentinel-2 surface reflectance and the corresponding UAV reflectance in the blue, green, red, red-edge (RE), and NIR bands for the uncorrected and the radiometrically corrected UAV imagery based on MOR, PIF, MAD-RRN, SVR-RRN, and MACA.

of 36%–82% and a reduction in RMSE of 89%–95%. Conversely, the implementation of MOR did not show a significant improvement but resulted in an obvious overestimation in most bands. After the MOR correction, the RMSE values decreased

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TABLE III
STATISTICAL ASSESSMENT OF VIS FROM SENTINEL-2 AND THE UAV IMAGERY BEFORE AND AFTER THE RADIOMETRIC CORRECTIONS

		Uncorrected	MOR	PIF	MAD-RRN	SVR-RRN	MACA
NDVI	R^2	0.759	0.512	0.801	0.811	0.727	0.837
	RMSE	0.186	0.184	0.044	0.041	0.055	0.037
CIgreen	R^2	0.610	0.599	0.829	0.846	0.745	0.865
	RMSE	3.332	1.605	0.313	0.280	0.376	0.260
CIre	R^2	0.581	0.514	0.705	0.713	0.721	0.843
	RMSE	0.427	0.209	0.068	0.077	0.059	0.043
MTCI	R^2	0.003	0.002	0.555	0.559	0.613	0.802
	RMSE	3.669	4.598	0.115	0.129	0.080	0.052
ENDVI	R^2	0.747	0.485	0.833	0.835	0.743	0.862
	RMSE	0.235	0.113	0.028	0.026	0.035	0.024
EVI	R^2	0.570	0.024	0.778	0.805	0.719	0.825
	RMSE	0.359	1.211	0.065	0.058	0.081	0.054

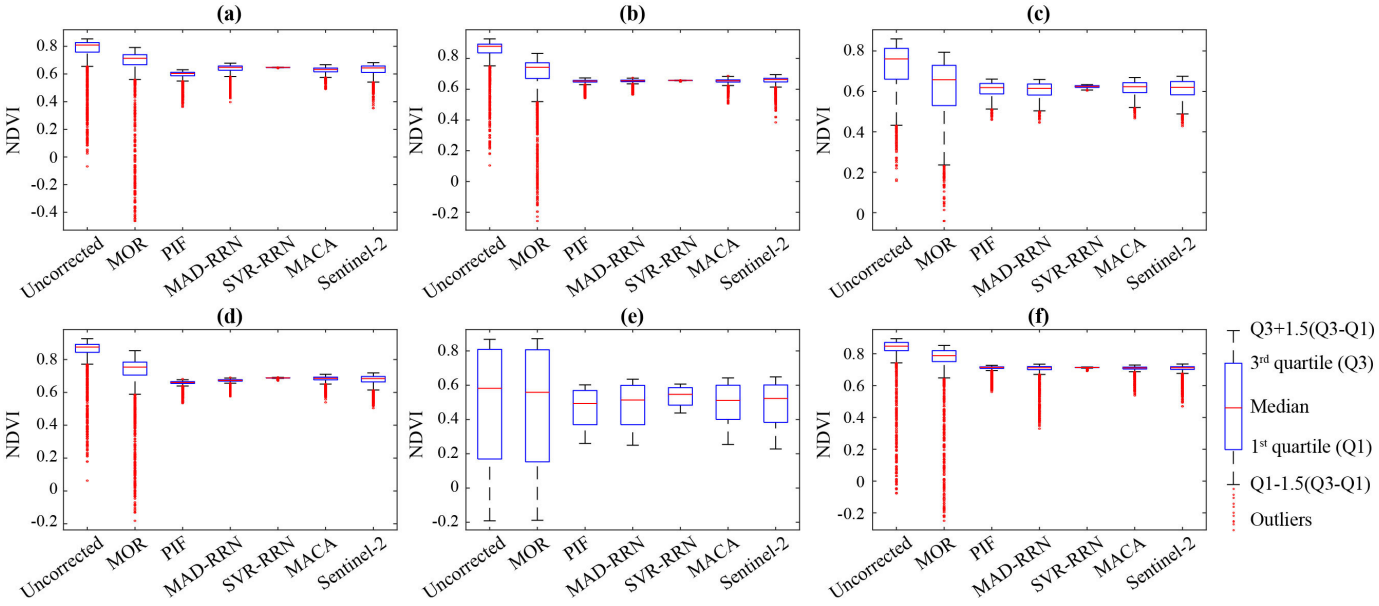


Fig. 5. Boxplots of the NDVI from the UAV imagery before and after the radiometric corrections, and the Sentinel-2 imagery at (a)–(f) sites 1–6.

in the green and blue bands and slightly increased in the red band with the data points for all bands being stretched horizontally. In contrast, PIF, MAD-RRN, and SVR-RRN had better performance with higher R^2 and lower RMSE than MOR in all bands. However, the accuracies of PIF, MAD-RRN, and SVR-RRN in the RE band were lower ($R^2 = 0.49$–0.53 and RMSE $= 0.013$–0.014) (see Fig. 4) than that of the MACA method ($R^2 = 0.81$ and RMSE $= 0.008$). Moreover, the reflectance ranges of SVR-RRN-corrected imagery for each site were relatively narrow.

B. Evaluation of Correction for VIs

As shown in Table III, the MACA had the best performance for all VIs with the highest R^2 (0.802–0.865) and the lowest RMSE (0.024–0.260). The performance of PIF (R^2: 0.555–0.833, RMSE: 0.028–0.313) and MAD-RRN (R^2: 0.559–0.846, RMSE: 0.026–0.280) was similar. Compared with the other methods, the SVR-RRN performed only better than MOR, which had the lowest accuracy (R^2: 0.002–0.599, RMSE: 0.113–4.598).

Taking the performance of NDVI as an example, the ranges of NDVI from the uncorrected UAV imagery were wider than that of Sentinel-2 NDVI for each site, with more outliers in the lower end (see Fig. 5). After the MOR correction, the NDVI values were reduced but the NDVI range was not narrowed. Conversely, the NDVI dispersions of the PIF, MAD-RRN, and MACA-corrected UAV imagery were similar to that of Sentinel-2 NDVI. However, the NDVI values of the PIF-corrected UAV imagery were slightly lower than Sentinel-2 NDVI values. Similar to the results of spectral bands in Fig. 4, the ranges of NDVI from the SVR-RRN-corrected UAV imagery were significantly narrow [see Figs. 5 and 6(e)]. Generally, the distribution of NDVI from the MACA-corrected UAV imagery was closer to that of Sentinel-2 NDVI for each site. As such, the MACA yielded the highest R^2 (0.84) and the lowest RMSE (0.037) for all

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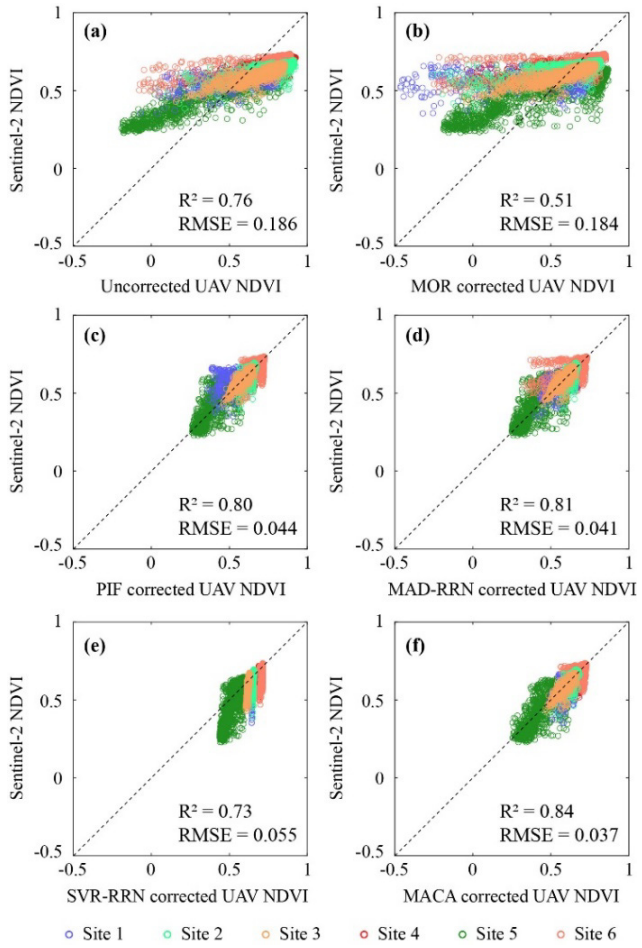


Fig. 6. Scatterplots of the Sentinel-2 NDVI and (a) NDVI from the uncorrected UAV imagery and the radiometrically corrected UAV-based NDVI using (b) MOR, (c) PIF, (d) MAD-RRN, (e) SVR-RRN, and (f) MACA.

sites (see Fig. 6). Similar patterns could be seen for other Vis, as shown in Table III.

C. Application of Corrected Imagery to PNA Estimation

Fig. 7 shows the correlation coefficients between VIs and the measured PNA for all study sites. Compared with the correlations from the uncorrected UAV imagery, the correlations from the MOR-corrected UAV imagery increased for CIgreen, MTCI, and EVI but decreased for NDVI, CIre and ENDVI. Similarly, the correlation for MTCI decreased from 0.227 to 0.120 after the PIF and MAD-RRN correction. In contrast, the VIs from the MACA-corrected UAV imagery had the highest correlations with PNA (Pearson correlation coefficient $>$ 0.58), followed by the SVR-RRN. Moreover, the random forest regression model developed using the VIs from the MOR-corrected UAV imagery had poor performance on the PNA predictions [see Fig. 8(b) and (h)]. Conversely, the model developed on the VIs from the MACA-corrected UAV imagery achieved the highest accuracy in both training [see Fig. 8(d)] and test datasets [see Fig. 8(h)].

With the corresponding random forest models, the PNA maps derived from different UAV imagery were generated, as shown in Fig. 9. The PNA map predicted from the MOR-corrected data did not show visible spatial contrast

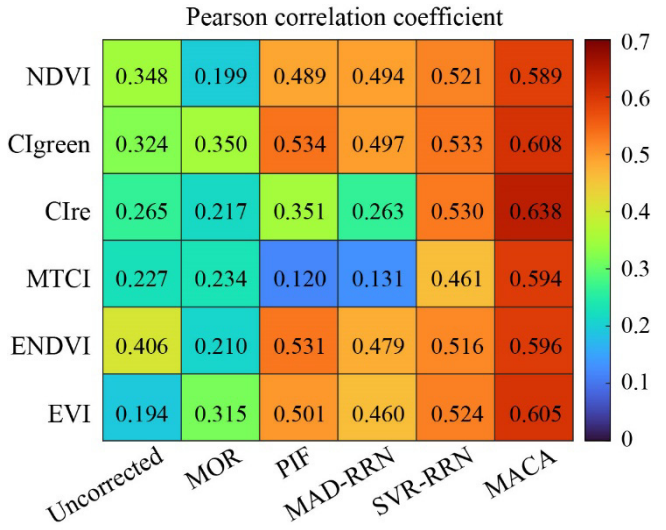


Fig. 7. Pearson correlation coefficient between the measured PNA at the six study sites and VIs extracted from the multiflight UAV images before and after relative radiometric correction using MOR, PIF, MAD-RRN, SVR-RRN, and MACA.

between fields and sites. While more heterogeneity in PNA was obtained after the PIF correction, there were some overestimations (e.g., at Site 2) and underestimations (e.g., at Site 6) of PNA compared with the filed measurements. Similarly, the lack of spatial details and underestimations were also observed in the PNA maps from the MAD-RRN (e.g., at Site 5) and SVR-RRN correction (e.g., at Site 2 and Site 5). As expected, the spatial pattern of the MACA-corrected PNA map could best show the differential distribution of PNA values within and between fields for all sites.

IV. Discussion

A. Advantages of MACA

While different radiometric calibration methods have been proposed, the absolute radiometric uncertainty of UAV imagery is likely to remain significantly larger than rigorously calibrated satellite data, such as Sentinel-2 and Landsat 8 [67]–[69]. MACA was proposed to deal with such potential radiometric uncertainties of multiflight UAV imagery based on rigorously calibrated satellite data (i.e., Sentinel-2 data in this study). Unlike the classical MOR and PIF methods, MACA employs a machine-learning technique (i.e., cubist) to create an interim UAV imagery as reference at the fine resolution and then conducts relative radiometric correction models for each band.

MACA has three main advantages.

1) MACA can eliminate the effect of spectral misalignments between different sensors. Generally, the closer the central wavelength and the spectral response function, the smaller spectral misalignment between sensors. According to our results, the offset in the spectral responses between the Sentinel-2 and the Airphen bands (see Fig. 2) would influence accuracies of MOR, PIF, MAD-RRN, and SVR-RRN (see Fig. 4), especially for the RE bands. Their large spectral difference in RE bands over the study sites could be attributed to the

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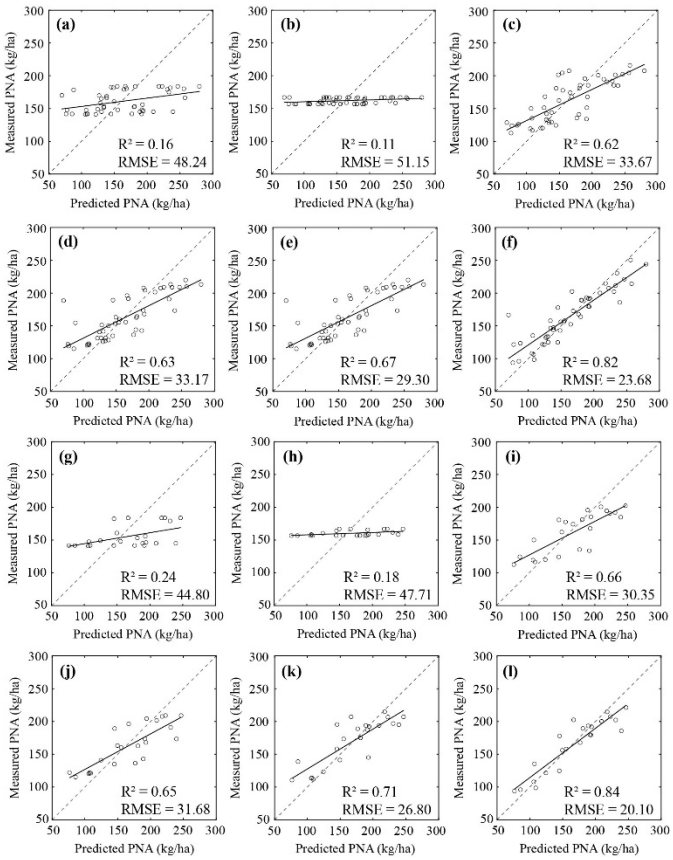


Fig. 8. Scatterplots of predicted PNA using random forest models developed on (a) and (g) uncorrected UAV data, and the UAV data corrected by (b) and (h) MOR, (c) and (i) PIF, (d) and (j) MAD-RRN, (e) and (k) SVR-RRN, and (f) and (l) MACA for (a)–(f) calibration based on training datasets and (g)–(l) validation based on test datasets.

variation in reflectance of vegetation across the RE region [70]–[72]. While other methods use a single band of UAV data similar to the satellite band, the CSF step in MACA employs all bands of UAV data as input covariates for training the model and optimizing the predictions. Hence, MACA is more robust than the classical methods in terms of the spectral mismatch effect. It indicates that the usage of multiple inputs can improve the performance of radiometric correction methods. Given this advantage, MACA has a great potential for different image pairs of UAV and satellite data even with spectral misalignments.

2) MACA is automatic and is not subject to errors in the selection of PIFs. For most applications, the selection of ground targets or PIFs is often subjective (e.g., PIF). While the MAD-RRN and SVR-RRN use automatic methods for features selection, they did not always perform better than the PIF. That is likely due to unavoidable errors caused by the automatic methods of classification or detection of invariant pixels, which further introduce uncertainty into the developed mathematical relationships [73]. Moreover, a limited number of PIFs cannot fully represent the variability and diversity in the scene and may only work well for the small areas used in the normalization procedure. Particularly, the support vector machine regression models trained on the limited number of invariant pixels leaded to the large bias of the SVR-RRN corrections. In contrast, MACA considers the spectral and spatial characteristics of the entire image. As such, the training dataset for MACA contains the larger range of samples. Although the methods using the full scene tend to be affected by outliers [15], the influence of outliers can be ignored when massive data (>9 million pixels in this study) is used. Given massive amounts of data from multiflight UAV and satellite images, machine-learning methods have a great potential in radiometric correction between UAV and satellite data.
3) MACA could reduce the spatial scale effect of UAV and satellite data. Most existing RRN methods are designed for satellite datasets with much smaller spatial difference than that between UAV and satellite data. The cross-sensor relationship in RRN methods was established based on the resampled UAV image at the same spatial resolution as satellite data. When RRN methods are applied to UAV radiometric correction, large-scale differences between UAV and satellite data (from 10 to 100 or even 1000 times) might cause block effects, especially for the heterogeneous landscapes 52], [74]. While other RNN methods directly apply the model trained at coarse resolution to the UAV image at fine resolution, MACA adopts relative radiometric correction models (i.e., the least squares regression technique in this study) to calibrate UAV spectral information at fine resolution based on the reference images generated by pretrained cubist models. Particularly, the Cubesat-enabled spatiotemporal enhancement method (CESTEM) [35] also uses cubist but regards the CubeSat image generated by cubist as the final-corrected output. In contrast, MACA uses the image generated by cubist as the reference image for the radiometric correction at fine resolution at the FSC step. By implementing radiometric correction at the fine resolution, MACA could enhance spatial details in the corrected UAV imagery. However, various landscapes (e.g., urban areas) and satellite data (e.g., Landsat or MODIS) should be evaluated in future research.

B. Benefits and Potentials of MACA for Multisensor Synergies

MACA has the potential to be used for radiometrically correcting UAV imagery from different cameras. Since a range of cameras have been used in UAV remote sensing [1], [2], the cross-sensor consistency will facilitate the comparison between different data sources and analysis results. As the number of sensors onboard UAVs grows, many customized sensor systems are designed but their calibration accuracies are seldom reported by the manufactures or in the literature [10]. While some instruments and software packages have been provided by the manufactures for radiometric correction, they are only applicable to a specific sensor, and some of them are even less accurate than empirical calibration. For example,

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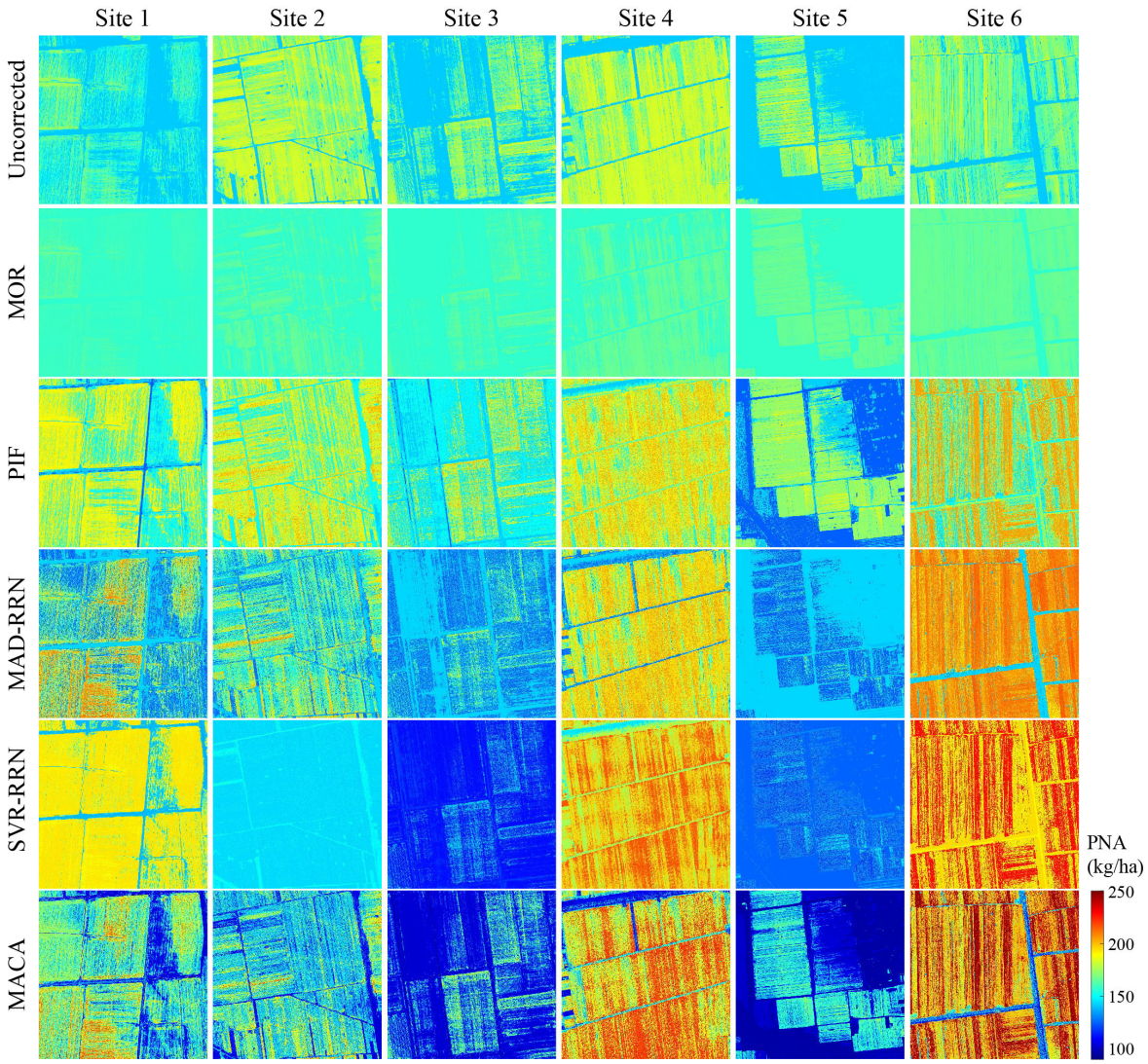


Fig. 9. Comparison of PNA maps derived from the UAV imagery before and after the radiometric corrections using the corresponding random forest models.

Jiang *et al.* [7] found that the empirical linear correction method using ground targets performed significantly better than the radiometric correction based on the incident light sensor initially equipped in the Tetracam mini-MCA6 camera. These limitations hinder the applications of multisensor data fusion for UAV-based crop monitoring. This study suggests that MACA can be used to cross-calibrate multisensor data into a standardized product, which will benefit multisensor synergies.

Moreover, MACA can enhance the synergy of UAV and satellite imagery for integrated multiscale crop monitoring. Specifically, UAVs often play a "bridging role" between *in situ* and satellite data in the synergy of UAV and satellite data [2]. Linking UAV and satellite data requires large amount of spatially matched pixels, especially when using machine-learning approaches. However, single-flight UAV acquisitions cannot provide wall-to-wall data over large spatial scales, which hinders their combination with satellite imagery with much wider swaths. An alternative strategy might be to use multiflight UAV datasets with several partial-coverage areas distributed within a satellite image (e.g., the six sites in this study), as this approach can cover a more complete representation of landscape features within a large area [75]–[77]. These studies [75]–[77] assumed the radiometric consistency of multiflight UAV data, which might result in prediction bias and reduced spatial variability (e.g., the uncorrected PNA mappings in Fig. 9). In this case, the satellite-calibrated UAV data obtained by MACA can be used to extend the *in situ* observations or models to larger scales.

C. Limitations of MACA and Future Research Directions

MACA would be readily affected by the spatial scale difference between UAV and satellite images. Large-scale differences between UAV and satellite images may result in greater geo-registration errors. Since MACA operates at pixel level, it is sensitive to the geo-registration errors of cross-sensor images at different spatial scales. The corresponding UAV and Sentinel-2 images were manually co-registered by selecting GCPs. However, automated geo-registration is highly recommended in future applications.

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For example, Houborg and McCabe [35] developed an automated co-registration method by maximizing the correlation coefficient between NDVI values from the reference and other images. However, their method was used to correct co-registration errors between the images at the same spatial resolution (e.g., CubeSat data in their study). Its capability to geo-reference the image pairs with large spatial scale differences (i.e., UAV and satellite data) needs to be further investigated. It would also be worthwhile to consider the scale invariant feature transform (SIFT) technique [78] for scene co-registration.

In addition, the temporal gap between UAV and satellite imagery may affect the accuracy of MACA and its applications. The temporal gap between UAV and Sentinel-2 used in this study is 0–1 day. While the PNA of winter wheat changes in the season, it could be considered invariant within one day. However, the ideally concurrent satellite imagery may be unavailable. As such, the prediction of PNA based on MACA-corrected data might result in bias, especially in the transition of different crop growth stages. In this case, daily satellite data (e.g., planet) or fused images might be considered as the reference. Basically, MACA can be improved by dealing with the issue of the temporal gap.

In this study, we only tested the performance of MACA in the estimation of PNA in winter wheat using UAV-based Airphen imagery. Given the high accuracy of spectral bands and VIs, MACA has a great potential for improving the estimations of other crop traits (e.g., leaf area index and chlorophyll content). More experiments based on different UAV data should also be evaluated for various crops under different growing conditions.

The MACA framework is highly generic and can be further improved by developing more alternative schemes for the CSF and FSC steps. In the CSF step, we used cubist due to its successful applications and outperforms compared with nonparametric machine-learning methods for regression purposes in numerous studies [35], [51], [79], [80]. However, with the rapid development of artificial intelligence technology, additional enhancements may be achieved by employing emerging machine-learning methods for improving computing efficiency and retrieval robustness. It is also worthwhile to investigate different RRN methods in the FSC step, such as nonlinear regression models or other regression strategies [15].

Due to the stable applicability and highly generic, MACA has a great potential in the application to other landscapes. However, it should be noted that MACA may not perform well in areas that cannot achieve high stability of illumination conditions during the image acquisition period. While satellite sensors capture a single image with a wide swath at one point in time, the UAV data are usually collected over a 15–20 min or even longer period for multiflight UAV data collection. As such, radiometric stability during the UAV data acquisition period is essential not only for MACA but also for other UAV relative radiometric correction methods with concurrent satellite imagery. The application of MACA in various landscapes should be evaluated in future research, especially in areas with persistent cloud cover, as well as urban areas with frequent tall and complex shadow-casting structures.

V. Conclusion

This article presents a new relative radiometric correction method, called MACA, for multiflight UAV images based on concurrent satellite data. MACA was evaluated using multiflight UAV imagery without overlapping areas and was compared with the MOR, PIF, MAD-RRN, and SVR-RRN methods. Compared with the benchmark methods, MACA could produce more consistent radiometric information between multiflight UAV and the concurrent Sentinel-2 data in spectral bands and VIs. An application of MACA produced substantial reduction in RMSE by up to 89%–95% and 80.1%–98.6% for spectral bands and VIs, respectively. Moreover, the application of MACA was beneficial to improve PNA estimations based on the corrected multiflight UAV data acquired at different sites and times. The models developed on the MACA-corrected UAV data achieved the highest accuracy in both training ($R^2 = 0.82$, RMSE = 23.68) and test datasets ($R^2 = 0.84$, RMSE = 20.10).

MACA has advantages in eliminating the effect of spectral misalignments between different sensors with all UAV bands as the inputs of cubist model and avoiding the selection of PIFs. In addition, MACA is easy to update by employing emerging machine-learning methods in the CSF step and other RRN methods in the FSC step. Furthermore, this method has a great potential in the application to multisensor synergies. The satellite-calibrated UAV data based on MACA could bridge the gap between filed and satellite observations and could allow better support to satellite-based crop monitoring by conducting unified models at fine resolution and extending *in situ* observations and the models to large scales. The standardized multisensor data products generated with MACA offer significant potential for improving the monitoring and characterization of terrestrial ecosystems at multiple scales.

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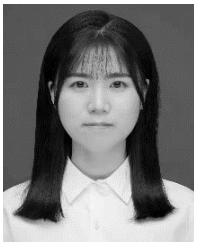
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